

Institutional Asset Allocation Engine

Modern Portfolio Optimization Report | August 04, 2026

Methodology

This allocation engine combines **Markowitz portfolio optimization** with the **Black-Litterman model**. The Markowitz frontier identifies efficient risk-return tradeoffs; Black-Litterman blends market-implied returns with subjective views to incorporate active insights. The final portfolio is optimized under practical constraints: no shorting, maximum 40% allocation per asset, and annual rebalancing.

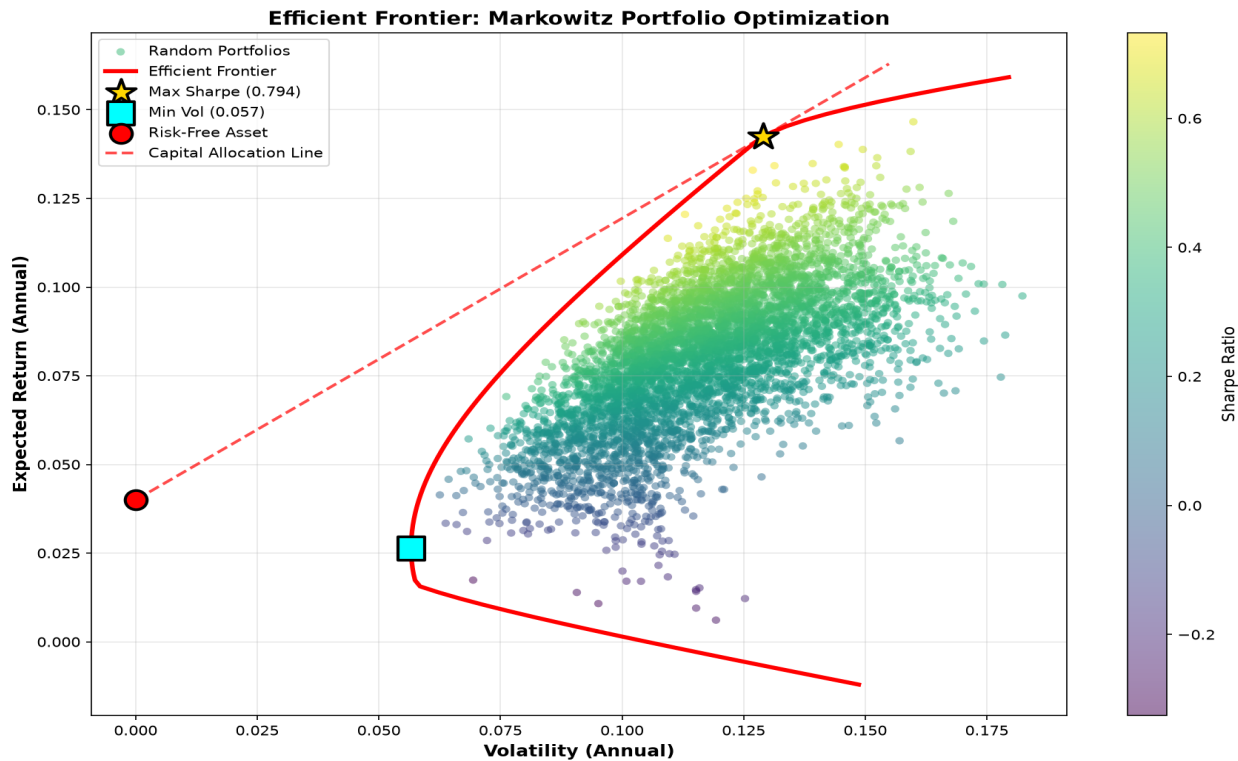
Asset Universe

Diversified portfolio of 7 ETF proxies: AGG, EEM, EFA, GLD, SPY, TLT, VNQ. Covers equities (US, international, emerging), fixed income (Treasuries, investment-grade bonds), alternatives (gold), and real estate (REITs).

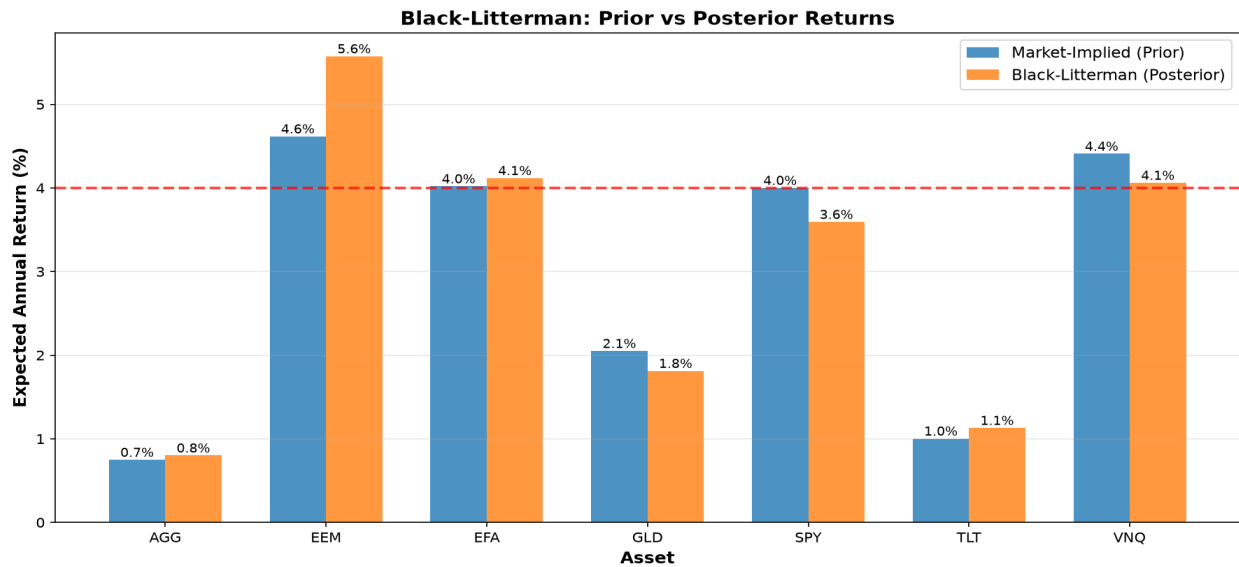
Subjective Views Incorporated

- EEM outperforms SPY by 2.0% (confidence: 50%)
- GLD outperforms AGG by 1.0% (confidence: 40%)

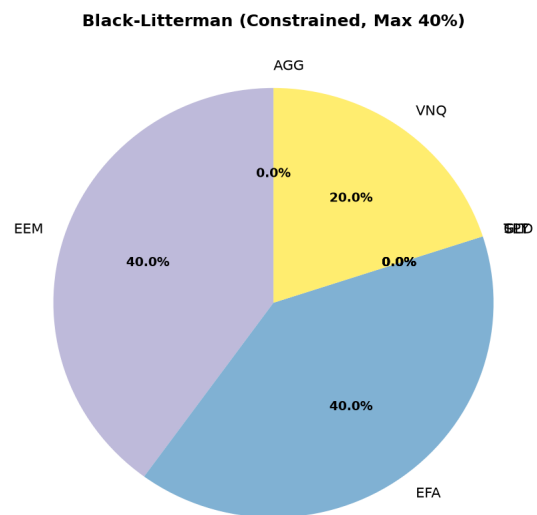
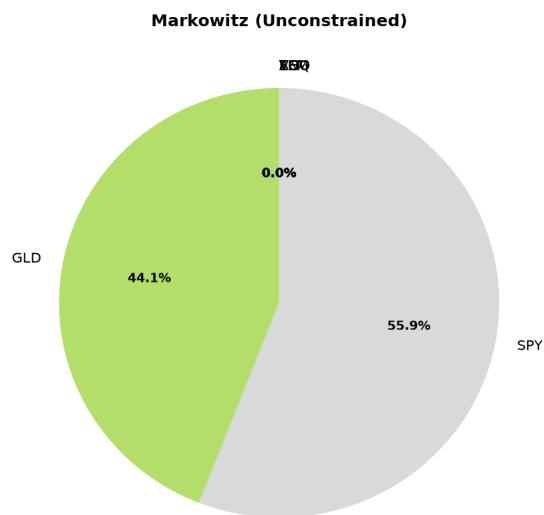
Efficient Frontier & Random Portfolios



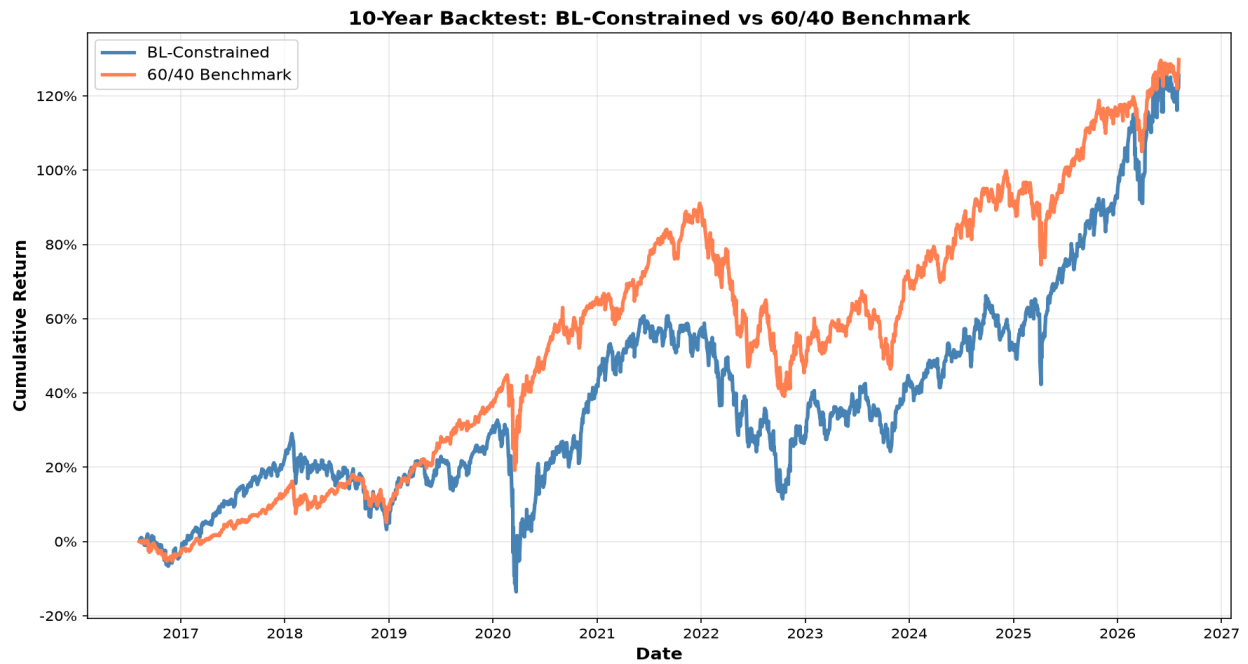
Black-Litterman: Prior vs Posterior Returns



Portfolio Allocations



10-Year Backtest Performance (Annual Rebalancing)



Metric	BL-Constrained	60/40 Benchmark
CAGR	8.49%	8.69%
Annual Volatility	7.85%	7.85%
Sharpe Ratio	0.571	0.597
Max Drawdown	-34.90%	-27.24%

This report is for informational purposes only and does not constitute investment advice. Past performance is not indicative of future results.